Test And Measurement For RUGGED APPLICATIONS

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Powerful portable workstations are needed for test and measurement purposes for rugged applications. This article takes a look at some such instruments

Professionals often face the challenge of capturing data in real time. For this, they need a portable solution to deploy in the field. The device must have massive storage and high-speed write performance capabilities. Hence, there is a need for powerful portable workstations for test and measurement purposes for rugged applications.

To withstand severe test cells and outdoor environments—from rain, sleet, snow and mud to off-highway vehicles and aircraft wings—electromechanical systems and components need thorough validation testing using reliable hardware.

Rugged applications deal with high temperature, pressure, hard usage and so on, depending on application. Precision and accuracy of components used in the test setups play a major role. These applications also depend on the need for real-time output.

Testing in rugged environments

It is important to plan while building monitoring or control systems with extreme temperature ranges for hazardous conditions. Otherwise, it can become difficult and expensive to obtain reliable measurements. While developing for rugged applications, designers must consider environmental certifications, temperature, liquid protection, shock, vibration and form factor of the hardware used in extreme and demanding environments.

As technology advances, the amount of data required to understand physical systems also increases, along with the need to accurately correlate different measurements. With more complex applications, sensor and synchronisation wiring has become increasingly difficult. An alternative approach is to distribute data acquisition (DAQ) devices around applications. Use of time-based synchronisation on all measurements from different devices helps integrate time-sensitive networking (TSN) technology into distributed measurement systems, and run a single network cable for data transfer back to the control room.

Features required in T&M devices

Rugged devices should be dust- and water-resistant, and TSN-enabled for simplified distribution into rough environments. These devices should allow accurate measure-

Fig. 1: Test centre for a chemical industry (Credit: www.gea.com)